A Novel Approach to Characterize Data Uncertainty in Material Flow Analysis and its Application to Plastics Flows in Austria

David Laner, Julia Feketitsch, Helmut Rechberger, and Johann Fellner

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Supporting information is available on the *JIE* Web site

Summary

Material flow analysis (MFA) is widely used to investigate flows and stocks of resources or pollutants in a defined system. Data availability to quantify material flows on a national or global level is often limited owing to data scarcity or lacking data. MFA input data are therefore considered inherently uncertain. In this work, an approach to characterize the uncertainty of MFA input data is presented and applied to a case study on plastics flows in major Austrian consumption sectors in the year 2010. The developed approach consists of data quality assessment as a basis for estimating the uncertainty of input data. Four different implementations of the approach with respect to the translation of indicator scores to uncertainty ranges (linear- vs. exponential-type functions) and underlying probability distributions (normal vs. log-normal) are examined. The case study results indicate that the way of deriving uncertainty estimates for material flows has a stronger effect on the uncertainty ranges of the resulting plastics flows than the assumptions about the underlying probability distributions. Because these uncertainty estimates originate from data quality evaluation as well as uncertainty characterization, it is crucial to use a well-defined approach, building on several steps to ensure the consistent translation of the data quality underlying material flow calculations into their associated uncertainties. Although subjectivity is inherent in uncertainty assessment in MFA, the proposed approach is consistent and provides a comprehensive documentation of the choices underlying the uncertainty analysis, which is essential to interpret the results and use MFA as a decision support tool.

Introduction

Material flow analysis (MFA) is about gathering, harmonizing, and analyzing data on physical flows and stocks (Brunner and Rechberger 2004). Regional MFA can be used to observe resource flows and trace resource losses from a defined system (i.e., cities, states, and nations) into the environment (Graedel et al. 2004; Modaresi and Müller 2012; Ott and Rechberger 2012). To quantify the resource and material flows, different data sources have to be taken into account, including official statistics, published and unpublished reports, digital sources, and expert's estimates. Each date of these different sources is associated with an individual uncertainty; for instance, an expert's estimate can be influenced by question formulation, company policy, or political opinion. To date, uncertainty of MFA input data has often been neglected or poorly described in MFA studies, thereby impairing statements about the reliability of MFA results for decision support (Laner et al. 2014).

Approaches for handling uncertainty within MFA range from qualitative discussions of the results' reliability to complicated mathematical models building on probabilistic methods for uncertainty analysis. Based on Laner and colleagues

Address correspondence to: David Laner, Vienna University of Technology, Karlsplatz 13/226, 1040 Vienna, Austria. *Email*: david.laner@tuwien.ac.at; *Web*: http://iwr.tuwien.ac.at/ressourcen/

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(2014), three groups of uncertainty approaches can be distinguished: (1) qualitative and semiquantitative approaches; (2) approaches based on data classification; and (3) statistical approaches. Qualitative and semiquantitative approaches are used to express our confidence in the MFA results without formally characterizing the uncertainty of the input data. Data classification approaches typically focus on the evaluation of input data quality (e.g., Weidema and Wesnæs 1996), but do not include mathematically rigorous procedures for propagating uncertainties through the model, whereas statistical approaches also propagate uncertainty through the material flow model to derive consequent uncertainty estimates for the model results. However, irrespective of the level of sophistication of an approach, the lack of information about MFA input data was identified as a major challenge for meaningful uncertainty analysis (Laner et al. 2014). Because the material flows are specific to a certain geographical frame, time period, and the good or substance, appropriate data for the actual system under investigation may often not be available (Do et al. 2014; Wu et al. 2014). Therefore, data corresponding to the quantity of interest are derived from up- and downscaling, based on transformation of data reported for other (similar) systems, or using expert's judgment. In order to describe the quality of the data in view of the actual quantity of interest and to consequently acknowledge data gaps, data quality assessment is used as a basis for characterizing the uncertainty of MFA input data. The rigorous characterization and documentation of uncertainty in MFA has several advantages; in particular, it (1) makes the definition of uncertainty transparent and enables reproducibility, (2) facilitates the creation of more reliable databases for MFA, (3) generates an understanding of the robustness of MFA results for decision support, and (4) allows for reconciling inconsistent data sets with the mass balance constraints of the model (Laner et al. 2015). Therefore, comprehensive data quality assessment and uncertainty characterization are important for profound uncertainty analyses, which lead to a better understanding of the reliability of MFA results and of the need to improve underlying material flow data.

It is the aim of this article to develop an approach to characterize the uncertainty of MFA input data and show alternative implementations of this approach using the Austrian plastics budget for the year 2010 as a case study. The analysis of plastics flows in Austria in 2010 (= Austrian plastics budget in 2010) is selected as a case study because of the following facts: Global plastics consumption has increased by 9% per annum since 1950 and currently accounts for more than 250 million metric tonnes (Plastics Europe 2012), because of the low material weight, low production cost, and multifunctional applicability of plastics. The different applications and increasing consumption of plastics, together with the large variety of the material types owing to the numerous types of polymers and additives used, lead to large, highly diverse plastics material flows in the anthroposphere. In order to illustrate the uncertainty characterization approach, a subsystem of the Austrian plastics budget focusing on the plastics flows in the major consumption sectors is used. The data available to quantify plastics flows into and out of the use stage vary widely with respect to sources and quality, which is why this subsystem is chosen as a showcase.

Material and Methods

Novel Approach for Uncertainty Characterization of Material Flow Analysis Data

The uncertainty characterization approach builds on an assessment of data quality following the scheme presented by Weidema and Wesnæs (1996) for life cycle inventory (LCI) data, on the one hand, and on the assessment of material flow data uncertainty using data classification, as proposed by Hedbrant and Sörme (2001), on the other hand. The so-called pedigree matrix consists of five independent data quality indicators (reliability, completeness, temporal correlation, geographical correlation, and further technological correlation) and was developed by Weidema and Wesnæs (1996) to characterize uncertainties of LCI data. Based on an evaluation of the data quality with respect to each indicator, corresponding scores are derived and used to calculate an uncertainty estimate for each date. Data uncertainties are expressed by coefficients of variation (CVs; standard deviation divided by mean) and can subsequently be used to calculate the effect of data uncertainty on the results of the study. The pedigree approach has also been implemented in the ecoinvent Database (Frischknecht and Jungbluth 2007) by using uncertainty factors instead of CVs. The uncertainty factors contribute directly to the quantitative uncertainty of a date, which is expressed as the geometric standard deviation. The general advantage of this approach is that data quality can be directly used to estimate the quantitative uncertainty of the respective inventory data and the effect of these uncertainties on the results. The approach for estimating uncertainty ranges by Hedbrant and Sörme (2001) was developed to study heavy metal flows in urban systems. They tried to quantify typical uncertainties within their MFA, which are "very likely" estimations of experts of the 95% confidence interval, based on a classification of the input data according to the individual data sources. According to their concept, large uncertainties are translated into asymmetric uncertainty intervals with the lower and upper bound defined by division or multiplication of the most probable value with an uncertainty factor, respectively. Hedbrant and Sörme (2001) distinguish five uncertainty levels that correspond to different sources of data and are used to calculate the uncertainty factors applying a continuous function. Owing to the nature of the uncertainty ranges, they cannot result in negative mass flows, which is a desired property in the case of characterizing MFA data (i.e., negative material flows invert the flow direction and therefore contradict the material flow model [Laner and Cencic 2013], with the only exception of stock changes, which can be positive or negative). However, whereas the data classification approach is used to derive asymmetric intervals, Hedbrant and Sörme (2001) assume the intervals to have qualities similar to the standard deviations of normal distributions for performing calculations. Whereas this is justified for small uncertainty factors, it gives rise to the

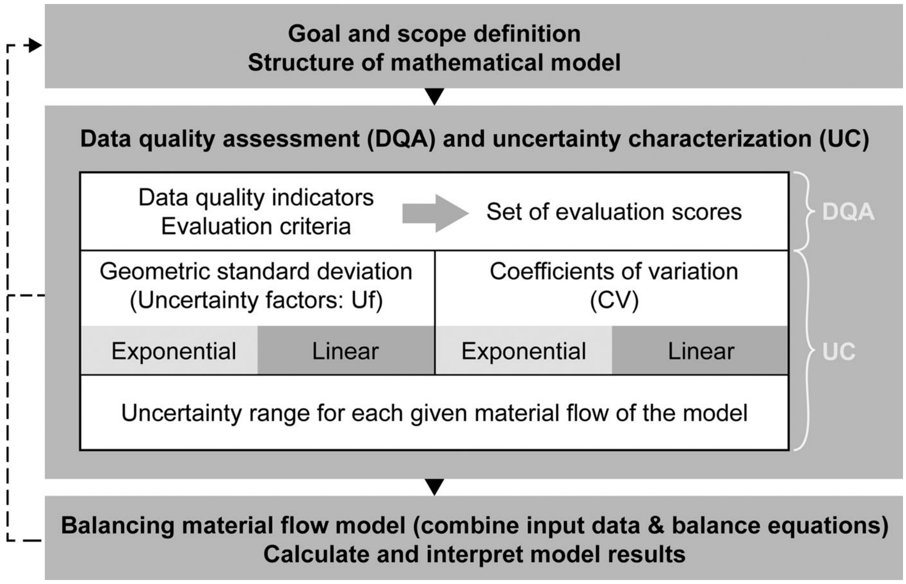

Figure 1 Approach for data quality assessment using data quality indicators and uncertainty characterization (testing four alternative implementations) of input data of the Austrian plastics budget.

mathematical inconsistency in the case of larger uncertainty factors (Limpert et al. 2001). Whereas the above-described concepts deal with some specific aspects relevant for uncertainty characterization in MFA, only their combination allows for a comprehensive uncertainty assessment. For instance, the data classification approach by Hedbrant and Sörme (2001) addresses the data source, but needs to be extended to also address the representativeness of data with respect to the quantity of interest. This is done by using data quality indicators, such as proposed within the pedigree concept (Weidema and Wesnæs 1996), and the introduction of additional indicators for the evaluation of expert estimates, which are often used if published data or measurements do not exist. In addition, extensions concerning the sensitivity of the quantity of interest with respect to specific aspects of data quality or concerning the definition of different functions to derive uncertainty estimates from data quality evaluation are introduced. The details of the developed approach and its implementation are described below.

The method builds on data quality indicators according to the pedigree scheme, but does not include a basic uncertainty factor, because it is assumed that there is a true value describing the quantity of interest for the system that is not known and needs to be estimated using imperfect data. For the quantification of uncertainty, different functions are defined to express how sensitive a date is with respect to a deviation in a specific aspect of data quality (i.e., an indicator score), building on the continuous functions suggested by Hedbrant and Sörme (2001). Subsequently, uncertainty estimates are derived for all material flows using different characterizing functions. In the first case, all uncertain quantities are assumed to be normally distributed given by mean value and standard deviation. In the second case, uncertainty ranges are defined by multiplication or division with an uncertainty factor. The uncertainty factor is equal to the geometric standard deviation of a log-normal distribution with identical arithmetic mean. The resulting asymmetric uncertainty interval corresponds to the width of a 68.3% confidence interval with respect to the median (= geometric mean).

In figure 1, the major steps of the proposed approach for data quality assessment and uncertainty characterization are illustrated. The method is put into context with the general framework of uncertainty analysis in MFA by Laner and colleagues (2014), where it corresponds to the second step of data uncertainty characterization. The first step of establishing the mathematical model and the subsequent steps of balancing the model, model calibration, and, potentially, sensitivity and scenario analysis are not dealt with in this work. Here, the focus is on the uncertainty characterization of MFA data, which comprises the definition of relevant indicators, formulation of criteria for assigning indicator scores based on the properties of a date, and quantification of uncertainties using specific characterizing functions. Four different ways of translating indicator scores into quantitative uncertainty estimates (so-called implementations of the approach) are tested. One pair of implementations expresses uncertainty as CV assuming normal distributions and differ with respect to the functions (exponential vs. linear type) used for determining the individual CVs associated with specific scores. The other pair uses uncertainty factors (corresponding to the geometric standard deviation) to express uncertainty assuming log-normal distributions (cf., figure 1). The overall concept and the various implementations are described in detail below.

Similar to the pedigree matrix of Weidema and Wesnæs (1996), five data quality indicators and criteria for assigning evaluation scores are defined. The indicators form the basis to link the quality of the data underlying the estimate of a material flow directly to the quantitative uncertainty of the flow. The

Table 1 Definition of data quality indicators and qualitative evaluation criteria for the application of score 1 to 4

Indicator	*Definition*	*Score: 1*	*Score: 2*	*Score: 3*	*Score: 4*
Reliability	Focus on the data source: documentation of data generation, e.g., assessment of sampling method, verification methods, reviewing processes.	Methodology of data generation well documented and consistent, peer-reviewed data.	Methodology of data generation is described, but not fully transparent; no verification.	Methodology not comprehensively described, but principle of data generation is clear; no verification.	Methodology of data generation unknown, no documentation available.
Completeness	Composition of the date of all relevant mass flows. Possible over- or underestimation is assessed.	Value includes all relevant processes/flows in question.	Value includes quantitatively main processes/flows in question.	Value includes partial important processes/flows, certainty of data gaps.	Only fragmented data available; important processes/mass flows are missing.
Temporal correlation	Congruence of the available date and the ideal date with respect to time reference.	Value relates to the right time period.	Deviation of value 1 to 5 years.	Deviation of value 5 to 10 years.	Deviation more than 10 years.
Geographical correlation	Congruence of the available date and the ideal date with respect to geographical reference.	Value relates to the studied region.	Value relates to similar socioeconomical region (GDP, consumption pattern).	Socioeconomically slightly different region.	Socioeconomically very different region.
Other correlation	Congruence of the available date and the ideal date with respect to technology, product, etc.	Value relates to the same product, the same technology, etc.	Values relate to similar technology, product, etc.	Values deviate from technology/product of interest, but rough correlations can be established based on experience or data.	Values deviate strongly from technology/ product of interest, with correlations being vague and speculative.

Note: GDP = gross domestic product.

indicators are *reliability*, *completeness*, *temporal correlation*, *geographical correlation*, and *other correlation* (see table 1). *Reliability* relates primarily to the data source and comprises the documentation of the data with respect to assessment of sampling method and the method used for data verification. The indicator *completeness* defines how complete the available date is in comparison to the date in question. It assesses possible over- or underestimations of the mass flow. These may be owing to missing processes or incomplete mass flows, for instance, when only a fraction of an input flow is known. The indicators *temporal* and *geographical correlation* describe the representativeness of the available date in comparison to the date of actual quantity of interest for the MFA (= ideal date) and describe deviations in time and space. The indicator *other correlation* takes other deviations into account besides from time and space, such as available data relating to a different product or technology than the one of the actual quantity of interest. Table 1 describes the correlation of the ideal date with the available data based on qualitative evaluation criteria specified for each indicator and each score. Score 1 would be assigned to a date that is fully congruent with the ideal date, whereas for score 4, a strong deviation between available date and ideal date exists.

Implementation of the Approach Using Normal Distributions

Subsequent to the data quality assessment, the quantitative uncertainty associated with a specific indicator score is determined. This is described for the characterization of uncertainty by CVs (assuming normal distributions) and the translation of indicator scores into quantitative uncertainty estimates based on an exponential-type continuous function (see figure S1 in the supporting information available on the Journal's website). Other implementations of the approach are described in the next section. The CVs presented in tables 2 and 3 correspond well with ranges previously estimated for LCI data, which range from a CV of 2% to 20% for very good data to a CV of 25% to 100% for very poor data (Weidema and Wesnæs 1996; Lloyd and Ries 2007). If factors were used to quantify uncertainty of input data, they were defined as 1.1 for the best data qualities and range from 10 to 100 to express the worst data qualities (Hedbrant and Sörme 2001; Lloyd and Ries 2007). Using factors, substantially larger ranges are often associated with very poor data quality, because there is no risk of negative estimates even for small quantities with large uncertainty, which would be problematic in the case of normal distributions. In any case, the definition of uncertainty ranges remains arbitrary to some degree, particularly if empirical data are not available to back up estimates. Therefore, consistency between the CVs relating to a specific indicator and a specific score is of crucial importance, which is assured in the method by the definition of continuous functions assigning quantitative uncertainties to different scores (equations 1 and 2). Depending on the sensitivity of the quantities of interest with respect to a deviation in a specific indicator, different parameterizations of the functions are available and can be chosen (table 2). For instance, the plastics content of electronic products is not sensitive to the indicator *geographical correlation* (data for Austria, Germany, Europe, or other countries are expected to relate to similar products), but may be sensitive to the indicator *temporal correlation* (data for products in 1990 may be very different from data for products in 2010 owing to technological progress and market changes). Thus, suggesting the CVs corresponding to the level "not sensitive," in the first case, and the CVs corresponding to the level "highly sensitive" in the latter case (table 2).

The CVs presented in tables 2 and 3 are based on continuous functions (see figure S1 in the supporting information on the Web), as suggested by Hedbrant and Sörme (2001) and defined in equation (1). In general, the choice of parameters is up to the modeler and depends on the specific data situation (see the *Discussion* section). Hence, the parameters are defined subjectively in a way so that the resulting uncertainty ranges of the flows are plausible and correspond reasonable to other MFA studies. The CV values are dependent only on two parameters for each evaluation score (1 to 4), and it is possible to calculate uncertainties also for intermediate scores (e.g., the relative uncertainty of 1.5 score is exactly defined). The parameter values for the different sensitivity levels are defined as follows: $a_{\text{not sensitive}} = 0.375$; $a_{\text{medium sensitive}} = 2*a_{\text{not sensitive}} = 0.75$; $a_{\text{highly sensitive}} = 4*a_{\text{not sensitive}} = 1.5$; $b_{\text{not sensitive}} = b_{\text{medium sensitive}} = b_{\text{highly sensitive}} = 1.105$.

$$\begin{aligned} &\text{For score} = 1 : \text{CV} = 0 \\ &\text{For score} = [1, 4] : \text{CV} = a \cdot e^{b\cdot(score-1)} \end{aligned} \tag{1}$$

The CVs for reliability are determined using a slightly adapted function (equation 2), because the best score should not result in zero uncertainty reflecting imperfect data generation in any case (cf., table 2). The parameter values used to calculate the values in table 2 are a = 0.75 and b = 1.105.

$$\text{For score} = [1, 4] : \text{CV} = a \cdot e^{b\cdot score} \tag{2}$$

After assigning quantitative uncertainty estimates (CVs) for each indicator, the overall uncertainty of a specific input date is determined by aggregating the individual CVs according to equation (3). The squared CVs are summed up to derive the squared coefficient of variation for the input date.

$$cv_{tot} = \sqrt{cv_{reliability}{}^2 + cv_{completeness}{}^2 + cv_{geogr.corr.}{}^2 + cv_{temp.corr.}{}^2 + cv_{other\ corr.}{}^2} \tag{3}$$

Table 2 Quantitative uncertainties for various sensitivity levels and indicators (shown for uncertainty characterization by coefficients of variation [CVs])

Data quality indicators	*Sensitivity level*	*Score: 1*	*Score: 2*	*Score: 3*	*Score: 4*
		CV (*in %*)			
Reliability	—	2.3	6.8	20.6	62.3
Completeness/temporal correlation/geographic correlation/other correlation	Highly sensitive	0.0	4.5	13.7	41.3
	Medium sensitive	0.0	2.3	6.8	20.6
	Not sensitive	0.0	1.1	3.4	10.3

Table 3 Coefficient of variation assigned to the reliability of an expert's estimate

Expert estimate	*Score: 1*	*Score: 2*	*Score: 3*	*Score: 4*
Evaluation criterion	Formal expert elicitation with (empirical) database – transparent procedure and fully informed experts on the subject	Structured expert estimate with some empirical data available *or* using transparent procedure with informed experts	Expert estimates with limited documentation and without empirical data available	Educated guess based on speculative or unverifiable assumptions
CV (in %)	4.5	13.7	41.3	124.6

Note: CV = coefficient of variation.

For instance, an evaluation of (1, 2, 3, 1, 2) for the amount of plastics-containing goods (highly sensitive to data quality deviations) would result in a total CV of 15.3% (CV_{tot} = sqrt($2.3^2+4.5^2+13.7^2+0.0^2+4.5^2$)). The overall CV is used to characterize the input data uncertainty in the material flow model, thereby identifying material flows with poor underlying data as highly uncertain. Consequently, material flows with large uncertainty ranges should be subject to further investigation in order to improve the database.

The procedure described above is designed for the case of published data sources, such as reports, studies, digital media, and so on, being available to estimate the quantity of interest. However, owing to frequent data scarcity, expert judgment is also an important source of MFA input data, which needs to be addressed by uncertainty characterization. Because expert estimates are solicited for the specific quantity of interest, an evaluation based on the *completeness*, *temporal*, *geographical*, and *other correlation* does not apply in this case. Instead, the reliability of the expert estimate is assessed according to the criteria shown in table 3. In essence, the quality of an expert estimate is based on the transparency and consistency of generating the estimate and the knowledge of the expert about the specific subject the estimate relates to. The CVs assigned to the individual evaluation scores are chosen in a way that the total uncertainty of the expert estimate is compatible with the corresponding aggregated uncertainty estimate (see equation 3) based on the data quality assessment (cf., tables 1 and 2).

Alternative Implementations Applied to the Case Study

In addition to CVs, uncertainties are alternatively characterized using uncertainty factors to allow for asymmetric uncertainty ranges and to acknowledge the non-negativity of material flow data (neither concentrations nor amounts of materials can be negative). Further, linear-type functions (see figures S2 and S4 in the supporting information on the Web) are implemented as an alternative to the exponential-type functions (see figures S1 and S3 in the supporting information on the Web) used to derive quantitative uncertainties for different indicator scores and sensitivity levels. In total, four different implementations of the approach (cf., figure 1) are investigated using the case study on plastic flows in Austria in 2010:

CV_exp: = Characterization of uncertainty by CVs (normal distributions) and translation of indicator scores into quantitative uncertainty estimates based on exponential-type functions.

U_f_exp: = Characterization of uncertainty by uncertainty factors (log-normal distributions) and translation of indicator scores into quantitative uncertainty estimates based on exponential-type functions.

CV_lin: = Characterization of uncertainty by CVs (normal distributions) and translation of indicator scores into quantitative uncertainty estimates based on linear-type functions.

U_f_lin: = Characterization of uncertainty by uncertainty factors (log-normal distributions) and translation of indicator scores into quantitative uncertainty estimates based on linear-type functions.

Uncertainty factors (U_f) are implemented alternatively to CVs assuming log-normal distributions for all uncertain quantities. To assure comparability of the concepts using the geometric standard deviation (factors, U_f) and the CVs, the relationship between CV and U_f is defined according to equation (4). The quantitative uncertainty estimates (uncertainty factors) are derived in the same way as described for the CVs. However, the aggregation of the uncertainty factor to calculate the overall uncertainty factor ($U_{f,tot.}$) for a material flow is done according equation (5). This is identical to the implementation of this concept in the ecoinvent Database for life cycle assessment (cf., Ciroth et al. 2013), with the exception that U_f here corresponds to the geometric standard deviation and not to its square (cf., equation 4). U_1 to U_5 relate to the individual data quality indicators, where U_1 is the uncertainty factor for *reliability*, U_2 for *completeness*, U_3 for *temporal correlation*, U_4 for *geographical correlation*, and U_5 for *other correlation*.

$$U_f = 1 + \left(\frac{CV}{100}\right) \tag{4}$$

$$U_{f,tot} = exp^{\sqrt{\ln(U_1)^2+\ln(U_2)^2+\ln(U_3)^2+\ln(U_4)^2+\ln(U_5)^2}} \tag{5}$$

Linear-type functions according to equation (6) are implemented alternatively to the exponential-type functions described above, to test another way of deriving uncertainty estimates based on the indicator scores. They are defined in equations (1) and (2) in a way that the evaluation scores of 1 and 4 result in the same values for both types of functions (for reasons of comparability). With respect to the various sensitivity levels, the corresponding parameters are: $k_{\text{not sensitive}} = 3.44$; $k_{\text{medium sensitive}} = 2{*}k_{\text{not sensitive}} = 6.88$; $k_{\text{highly sensitive}} = 4{*}k_{\text{not sensitive}} = 13.76$; $k_{\text{reliability}} = 20.02$; $k_{\text{expert estimate}} = 40.04$; $d_{\text{not sensitive}} = d_{\text{medium sensitive}} = d_{\text{highly sensitive}} = 0$; $d_{\text{reliability}} = 2.26$; $d_{\text{expert estimate}} = 4.53$.

$$CV = k \cdot (score - 1) + d \tag{6}$$

Because some material flows of the plastics budget are constituted by many individual flows (e.g., imports of plastics contained in various products) and because most material flows are derived by multiplying plastic contents and the amount of products (cf., tables S1 and S2 in the supporting information on the Web), further calculations are necessary to arrive at the material flows to be inserted into the MFA model.

For uncertainty ranges assuming normal distributions (CVs), the calculations are straightforward and can be performed according to the Gaussian law of error propagation (Baccini and Bader [1996]; also see equation (7) for addition or subtraction and equation (8) for multiplication or division). Where CV_A and CV_B are the CVs of the mean values of mass flow A (M_A) and mass flow B (M_B) and CV_C is the CV associated with the mean value of mass flow C (M_C).

$$\text{Addition or subtraction:}\quad \left(M_C * \frac{CV_C}{100}\right)^2 = \left(M_A * \frac{CV_A}{100}\right)^2 + \left(M_B * \frac{CV_B}{100}\right)^2 \tag{7}$$

$$\text{Multiplication or division:}\quad CV_C^2 = CV_A^2 + CV_B^2 \tag{8}$$

Asymmetric uncertainty ranges calculated by dividing or multiplying, respectively, with the overall uncertainty factors ($U_{f,tot}$) correspond to the 68.3% confidence interval with respect to the median (= geometric mean) of the log-normal distribution. Whereas the formula given in equation (5) can be directly used in the case of multiplication, the combination of overall uncertainty factors is not solved analytically in the case of additions or subtractions. For the latter, a log-normal distribution is defined for each mass flow, whereby the 15.85 percentile (geometric mean divided by geometric standard deviation) and the 84.15 percentile (geometric mean multiplied with geometric standard deviation) are derived by multiplication and division with the uncertainty factor $U_{f,tot}$. Calculations were computed by Monte Carlo simulation (MCS) with 10,000 runs for each individual material flow. From the resulting histogram, the 15.85 and 84.15 percentiles were used to derive the uncertainty factor associated with the material flow (median divided by 15.85 percentile or 84.15 percentile divided by the median). This final uncertainty factor is then used to quantify the uncertainty of the material flow inserted into the model (see tables S3 and S4 in the supporting information on the Web).

Material Flow Analysis of Plastics Flows in Austria in 2010

In order to determine potential stocks and losses (nonutilized flows) of plastics, all flows within the system are traced from cradle to grave and comprise the following processes: *primary production*, *foreign trade*, *preparation and manufacturing*, *consumption*, and *waste management*. The consumption sector is subdivided into nine subsectors and nonplastic applications, namely, Packaging, Building and construction, Transport, Electronic, Agriculture, Furniture, Medicine, Household, and Others (cf., figure 2). Waste flows are considered as outputs from the consumption sectors, but the waste management sector itself is not included in the model. This is owing to the focus of the case study on the plastics flows in the nine consumption sectors distinguished in the model (the sector Non-plastic applications is not further investigated, because these plastic flows cannot be recovered from wastes), for which inputs, outputs, and stock changes are evaluated using the developed uncertainty characterization concept.

Material flow modeling is performed with the software subSTance flow ANalysis (STAN) (freeware: www.stan2web.net/) which is a widely applied tool to perform MFA and establish material balances on a regional level (Cencic and Rechberger 2008). Mass balances are established based on information about the plastics contents of various goods and the amount of these goods produced, consumed, imported, and exported. Import and export data are retrieved from official statistics (Statistik Austria 2010). The plastics inputs into the different consumption sectors is estimated based on data reported for Germany (Lindner 2011), owing to the fact that no data about the sector split of plastics flows are available for Austria (cf., table S4 in the supporting information on the Web). Data for plastic waste flows are reported to a greater extent for sectors with waste-specific regulations in place, such as Packaging (ARA AG 2010), Transport (Kletzmayer 2011; WKO 2011), or waste electric and electronic equipment (WEEE) (Tesar and Öhlinger 2009; Elektroaltgeräte Koordinierungsstelle Austria GmbH 2010). For the other sectors, the amount of waste is

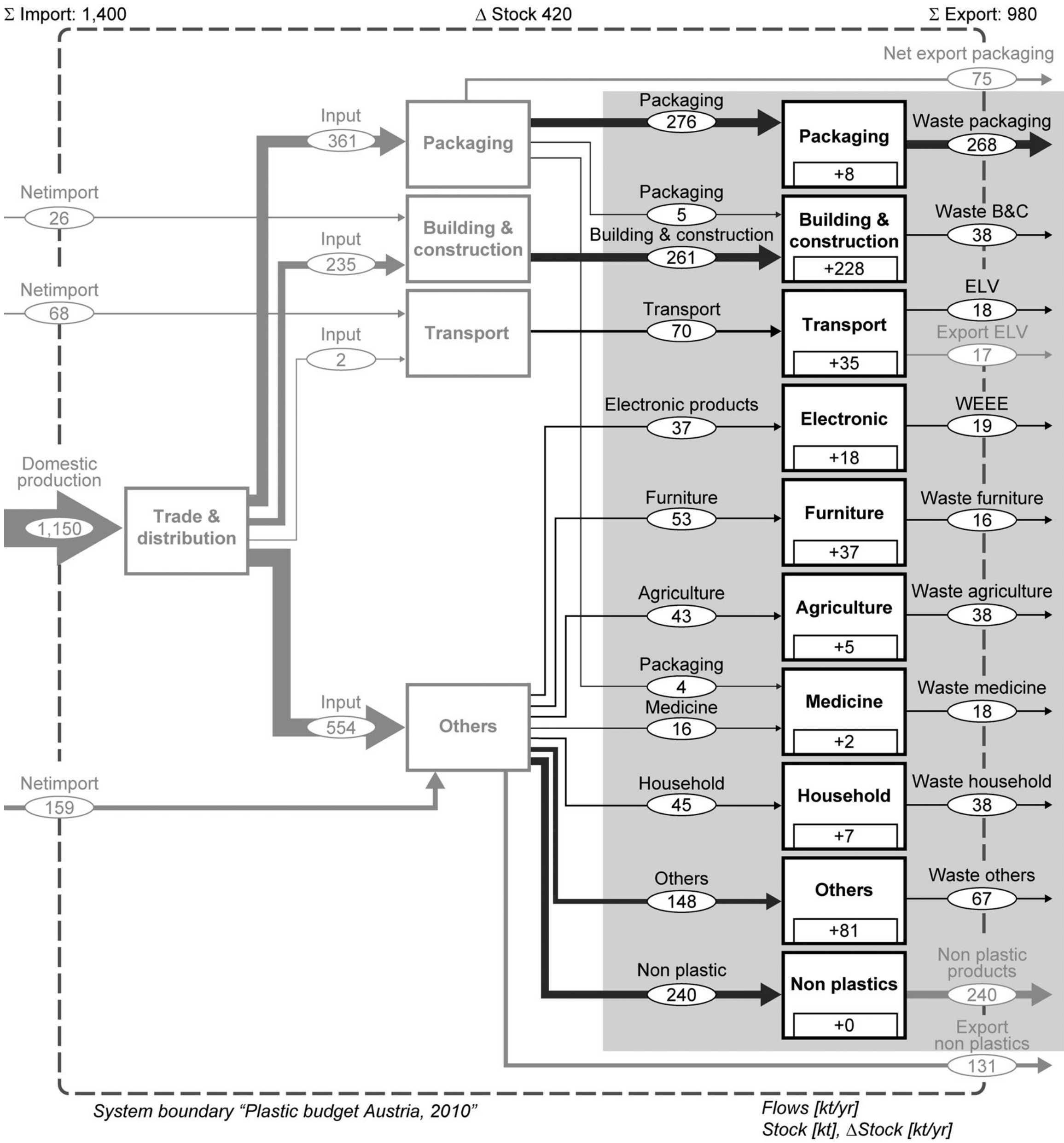

Figure 2 Sankey-style diagram of plastics flows in Austria in 2010 in kilotonnes per year (1 kilotonne = 1 Gg). The focus of the model lies on the flows into and out of the consumption sectors (shown as boxes in the gray area), whereby Non plastics is only listed for completeness, but not further investigated. The stock changes in each sector are shown within the boxes (positive numbers indicate stock growth). Gg = gigagram; kt/yr = kilotonne per year.

calculated as shares of different waste streams, such as furniture in bulky waste, plastics in construction and demolition waste, and so on. Data on the various waste streams originate mainly from the Austrian Federal Waste Management Plan (BMLFUW 2011). Plastic contents of diverse products are extracted from national and international studies (Fehringer 1997; Hutterer et al. 2000; Associations of Cities and Regions for Recycling 2004; Swedish Chemicals Agency 2004; Delgado et al. 2007; Sutter 2007; Obersteiner and Scherhaufer 2008; Vinyl 2010 2010; Scarascia-Mugnozza et al. 2011; Martinho et al. 2012). Despite these some reported data, the plastic contents in various products are treated as expert estimates in the case study, because the choice of an appropriate plastic content in a product is rather subjective given the poor data base.

Results

Plastics Flows in Austria in 2010

The most important production and consumption processes of plastics in Austria have been identified and included in the material flow model shown in figure 2. The amount of

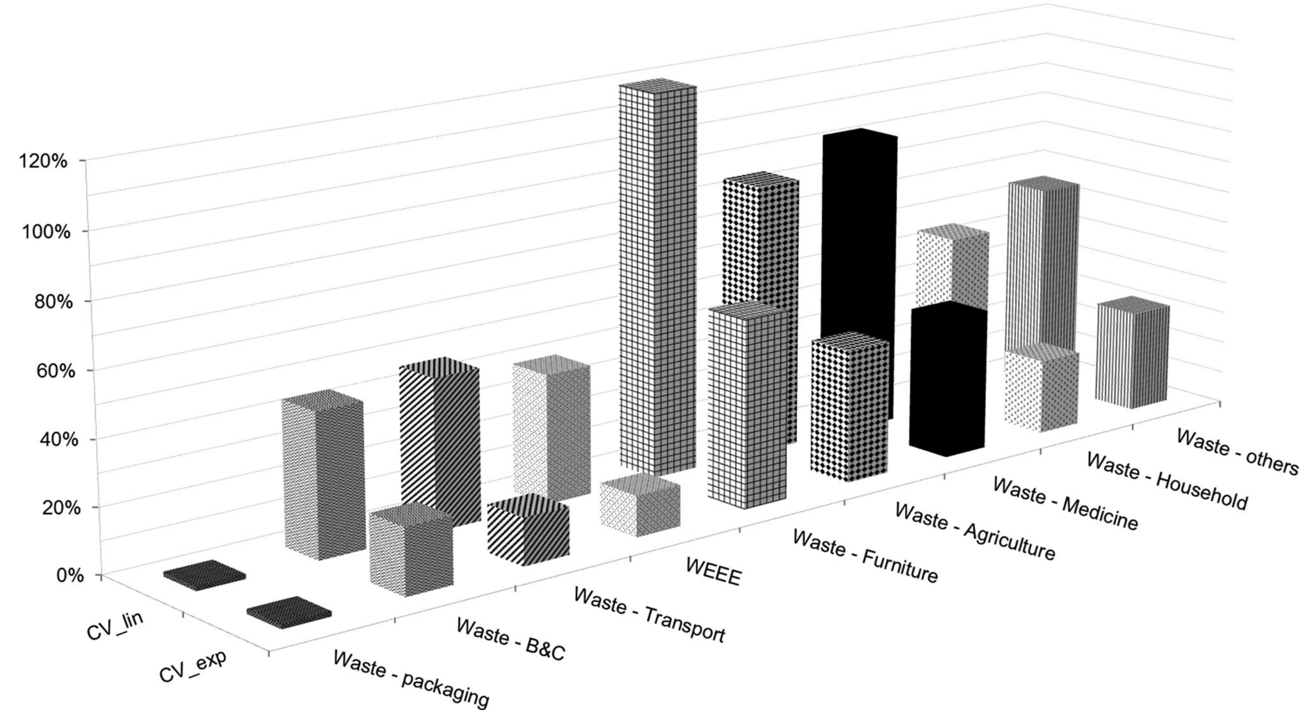

Figure 3 Uncertainty characterization results for plastics waste flows out of the nine major consumption sectors according to the implementations CV_exp and CV_lin. B&C = Building and construction; WEEE = waste electrical and electronic equipment.

plastics and semifinished goods produced in Austria in 2010 was 1,150 kilotonnes (kt). The total plastic consumption was 1,200 kt per year (kt/yr) (including imports and exports), which is equal to 130 kg per capita and year (kg/cap/yr) and in good agreement with average numbers for Europe and Germany (cf., Lindner 2011; PlasticsEurope 2012). The quantitatively most important consumption sectors are Packaging (30%), Building and Construction (18%) and Non plastic applications (20%), such as lacquers and adhesives. Using mass balancing, the stock changes were calculated for the major consumption sectors resulting in a total growth of the Austrian plastics stock of 420 kt in 2010, corresponding to a 50-kg increase per capita and year. This plastics stock increase is similar to the 60 kg/cap/yr reported by Lindner (2011) for Germany. The overall stock of plastics has not been determined in the present study. Postconsumer plastics wastes amount to 520 kt (= 62 kg/cap/yr), which is approximately 20% higher than the corresponding numbers for Germany (50 kg/cap/yr) (Lindner (2011).

Uncertainty Assessment of Plastics Flows in Major Consumption Sectors

In order to quantify the flows of plastics in Austria in 2010, various data sources of different quality had to be taken into account. For the plastics flows in the different consumption sectors, input data of highly diverse quality are available. Therefore, the developed approach, consisting of data quality assessment and uncertainty characterization, is illustrated for the material flows in the nine consumption sectors (gray box in figure 2). With respect to the availability and quality of plastics waste data, the situation is better for waste streams that are legally regulated by specific guidelines. This is reflected by the data quality assessment according to the developed approach, where average scores for waste streams from Packaging or Transport are 1.0 or 1.25, respectively, whereas they are 2.0 or above for waste furniture or construction and demolition waste (cf., table S2 in the supporting information on the Web). These differences in data quality are translated into the quantitative uncertainty estimates using exponential- or linear-type functions. The resulting uncertainties are shown in figure 3 for the implementations based on CVs to express uncertainty (normal distributions). The CV, and therefore the highest quality of the data, is associated with the plastics waste flow from Packaging (CV = 2%, both for CV_exp and CV_lin). Waste streams out of the sectors Transport (CVs are 15% and 47%, respectively) and Electronic (CVs are 13% and 40%, respectively) can be determined with relatively low uncertainty and are therefore well founded on data. For the other sectors, uncertainty ranges vary between 20% (Building and construction) and 58% (Furniture) in the case of CV_exp. For CV_lin, uncertainty ranges are substantially higher, ranging from 44% (Building and construction) to 120% (Furniture). The very high CV for plastics in furniture waste is owing to scarce data and assumptions underlying the determination of mass fractions (e.g., share of furniture in bulky waste) or the plastic content of furniture. Larger uncertainty ranges result from the application of the linear-type functions for uncertainty characterization, because indicator scores of 2 and 3 are translated into higher uncertainty

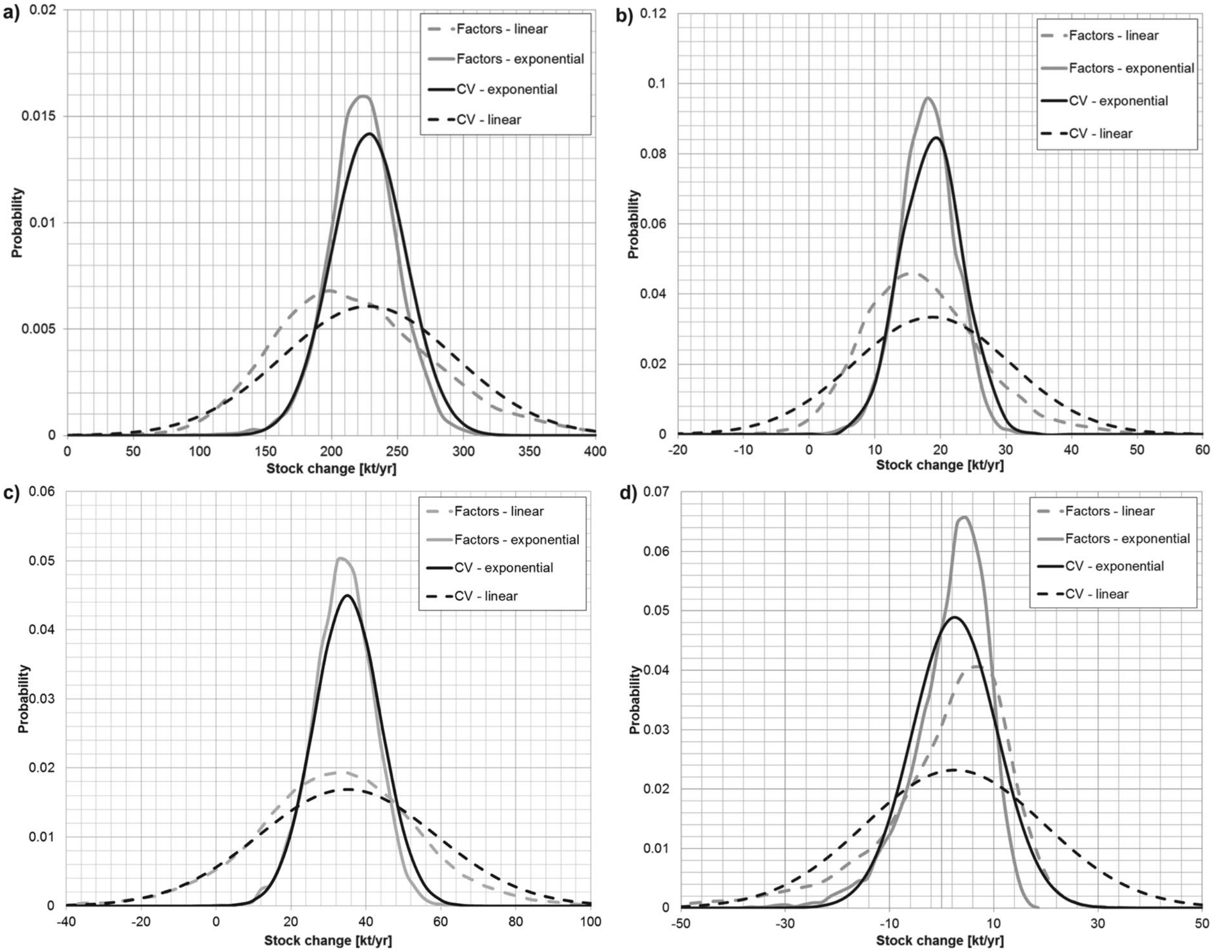

Figure 4 Calculated plastics stock changes based on the various implementations of the uncertainty characterization approach for four selected consumption sectors for Austria 2010 (a) Building and construction, (b) Electronic, (c) Transport, and (d) Medicine. Arithmetic means are identical independent of the implementation, but the probability density functions of the calculated stock changes differ depending on the uncertainty characterization scheme (CV_exp – black solid line, CV_lin – black dotted line, U_f_exp = grey solid line, U_f_lin = grey, dotted line). CV = coefficient of variation; kt/yr = kilotonne per year.

estimates. Hence, whereas uncertainty ranges for flows with very good (e.g., packaging waste) or extremely bad (average indicator score above 3) data quality are hardly influenced by the choice of exponential- or linear-type functions, flows with average to poor data quality are highly affected. However, it should be noted that although the values of the CVs for the individual waste flows are strongly influenced by the choice of function (linear or exponential), the ranking of the waste flows with the highest uncertainties is robust against the type of function. The same observations can be made if results from the Uf_exp and Uf_lin implementations are compared (cf., figure S5 in the supporting information on the Web). In essence, the waste flows with the highest and the lowest uncertainties are the same for both implementations. Thus, the uncertainty estimates derived by the various implementations of the developed approach deliver results consistent with the data available for calculating waste plastics flows.

Based on the determination of plastics flows into and out of the consumption sectors, the stock changes for each of the nine sectors are calculated by mass balancing (cf., figure 2). Whereas the arithmetic means of the plastics flows are the same for each implementation of the uncertainty evaluation approach, the resulting ranges and the shapes of the uncertainty estimates differ from one implementation to another. In figure 4, the changes of stocks resulting from closing the mass balances are shown for the consumption sectors Building and construction (figure 4a), Electronic (figure 4b), Transport (figure 4c), and Medicine (figure 4d). The resulting stock changes for the other five consumption sections are shown in figure S4 in the supporting information on the Web. In line with the findings outlined above, it is clearly visible from figure 4 that the implementations using the linear-type functions lead to substantially larger uncertainty ranges associated with the changes of stocks in the individual sectors. For the sector Building and construction, the 95% confidence interval for the median ranges roughly from 170 to 280 kt/yr, if exponential-type functions are implemented and from 100 to 360 kt/yr, if linear-type functions are used, assuming normal distributions (CV implementation). The corresponding values for the implementations using log-normal distributions (note: the probability density functions in figure 4 do not [necessarily] correspond to log-normal distributions, because they are derived from histograms calculated from mass balancing using MCS) range from 180 to 280 (U_f_exp) and from 120 to 370 kt/yr (U_f_lin), respectively. Thus, the effect of using different functions to translate data quality into quantitative uncertainties on the uncertainty of the stock change is more pronounced than the effect of either expressing uncertainty using normal or lognormal distributions. The difference in the results between the CV implementation and the U_f implementation is particularly small for the estimates with underlying exponential-type functions, because of the lower uncertainty estimates compared to the implementation using linear-type functions. However, the CV or U_f implementations have a marked effect on the uncertainty of plastics stock changes, for sectors with relatively high uncertainties if linear-type functions are used (dotted lines in figure 4). For instance, the 95% confidence interval for the median of plastics stock change in the sector Medicine ranges from –31 to 36 kt/yr based on CV_lin and from –36 to 21 kt/yr based on U_f_lin, respectively (see dotted lines in figure 4d). In this case, CV_lin results in a symmetric range centered at 2.4 kt/yr, whereas U_f_lin leads to an asymmetric range with the same arithmetic mean, but a median of 5.8 kt/yr (left-skewed), therefore giving higher probabilities to values above the mean. In general, the smaller the uncertainty, the less pronounced is the effect of assuming normal or lognormal distributions to characterize data uncertainty.

Discussion

There are two major aspects of subjectivity in the proposed uncertainty characterization approach, first, assigning indicator scores based on more or less stringent criteria, and second, the quantitative uncertainty estimates associated with different scores. Although the evaluation criteria do not leave much room for interpretation for some indicators (e.g., temporal correlation), there are others where the choice of scores (in particular, between scores 2 and 3) is not unambiguous, thereby allowing the modelers to make use of their own experience and express tacit knowledge. With respect to uncertainty characterization, explicit functions are used to define the relationships between different scores and quantitative uncertainty estimates. Whereas the use of defined functions to derive uncertainty estimates guarantees for a transparent and consistent characterization within the method, the definition of the parameter values (a and b for the exponential-type and k and d for the linear-type functions) are up to the modeler's choice. In this work, parameters are defined to correspond to uncertainty estimates established for other MFA studies (e.g., Ott and Rechberger 2012) and to ranges of the default uncertainty factors used in the ecoinvent LCI database (Frischknecht and Jungbluth 2007). Nevertheless, the actual uncertainties associated with material flows may be substantially different from one study to another because of different basic data and subjective assessments of the situation. For instance, flows of a substance mainly appearing as a minor constituent of products often being considered contaminant flows (e.g., heavy metals) (cf., Hedbrant and Sörme 2001) are typically associated with higher uncertainty than flows of a substance of economic interest present in greater amounts in the economy (e.g., aluminum) (cf., Buchner et al. 2014). Consequently, the translation of indicator scores into uncertainty estimates may differ from one MFA study to another, typically without the possibility of using empirical data as a basis for uncertainty estimation. Thus, whereas the proposed approach builds on reproducible, internally consistent uncertainty estimates, the comparison of uncertainty estimates generated in different MFA studies is not straightforward and should be done cautiously owing to some subjectivity involved in the uncertainty assessment in any case.

The four different implementations applied to the plastics flows in the major consumption sectors in Austria in 2010 result in different uncertainty ranges. Whereas the ranges are hardly influenced by the use of normal or log-normal distributions to characterize uncertainties in the case of small overall uncertainty, the ranges may differ substantially for flows associated with relatively high uncertainties. On the contrary, for the implementations using exponential-type functions and those using linear-type functions, pronounced differences in the resulting uncertainties can be observed independent of the assumed probability distributions. This is because many of the material flow data are associated with medium indicator scores (between 1 and 4), where the uncertainty estimates are strongly affected by the choice of either exponential- or linear-type curves (cf., figures S1 to S4 in the supporting information on the Web). Hence, the basic mechanism to generate uncertainty estimates (CVs or U_fs) plays a significant role with respect to the resulting uncertainty ranges of the material flows. By using specific continuous functions, however, it is assured that the resulting estimates are consistent relative to one another within a specific implementation (i.e., flows with the highest and lowest uncertainties are the same in any implementation). Such consistency is important because the relative uncertainties of material flows may have a potentially strong effect on data reconciliation within the mass balance equations or the evaluation of the most critical data underlying an MFA (cf., Dubois et al. 2014).

The advantage of assuming log-normal distributions (U_f implementations) for uncertainty characterization is that the associated variable is restricted to positive values only, which is typically the case for material flow data (concentrations, amounts of goods, and transfer coefficients). This is a particular advantage in the case of small quantities (close to zero) with large uncertainty ranges. If normal distributions are assumed for such flows, this can be problematic because flow values may be altered strongly during data reconciliation (owing to their high uncertainty) and thereby become negative, which then contradicts the physical material flow model (cf., Laner and Cencic 2013). However, the assumption of normal distributions (CVs) also has advantages, because error propagation can be analytically handled based on the Gaussian law of error propagation and data reconciliation can be performed by minimizing the sum of the squared errors. In addition, the concept is implemented in the widely used MFA software STAN, which makes it convenient to use for material flow modeling. In contrast, the use of log-normal distributions requires more resource-intensive MCS for aggregating flows and solving the mass balance equations (multiplications can be handled analytically; cf., equation 5). Overall, the assumption of normal distribution can be well justified and is convenient for small-to-moderate relative uncertainties (up to 30%) and if uncertainties do not exhibit strong intrinsic asymmetry (cf., Cencic and Frühwirth 2015). Otherwise, other probability or possibility distributions should be used, if their application is in better agreement with the underlying data or the level of information (cf., Laner et al. 2015).

Conclusion

In this article, a novel approach to characterize uncertainty of MFA input data in a transparent manner is presented. It is applied to assess the uncertainties of plastics flows in different Austrian consumption sectors in the year 2010. The approach consists of several steps, which can be summarized as follows:

1. Evaluation of data quality with respect to reliability, completeness, temporal and geographical correlation, and other correlations using indicator scores. For the evaluation of expert estimates, only one data quality dimension is used.
2. Translation of each indicator score into specific uncertainties, taking into account the sensitivity of the quantity of interest with respect to data quality imperfection.
3. Aggregation of the uncertainties associated with the individual scores to the overall uncertainty estimate for the material flow.

Within the case study, four different implementations for the translation of indicator scores to specific uncertainties (step 2) and their effect on the results are investigated. One pair of implementations expresses uncertainty as CV assuming normal distributions with two different functions (exponential vs. linear type) used for determining the individual CVs associated with specific scores. The other pair uses uncertainty factors (U_fs) assuming log-normal distributions. Whereas the assumption of normal vs. log-normal distribution only has a stronger effect on the plastics flows' uncertainty ranges in the case of high overall uncertainties, the use of exponential- vs. linear-type functions to derive uncertainty estimates has a substantial effect on the resulting ranges in any case. This indicates that the way of deriving uncertainty estimates for material flows may be more important than the assumptions about the underlying probability distributions. Because these uncertainty estimates originate from data quality evaluation as well as uncertainty characterization and because they are, to some degree, subjective in MFA, it is of crucial importance that this part of uncertainty analysis is done in a reproducible, consistent way. This is the reason for using a well-defined approach for uncertainty characterization, building on several steps to ensure the comprehensive translation of the data quality underlying material flow calculations into their associated uncertainties. In this way, the addressees of MFA results are equipped with a documentation of the rationales and calculations underlying the uncertainty analysis, which is essential to interpret the results and use MFA as a decision support tool.

The case study application indicates that the utilization of CVs do have several benefits in comparison to the use of uncertainty factors, primarily owing to the analytical solutions available for basic calculations and the use of the normal distribution assumption in the widely used STAN software for MFA. With respect to exponential- vs. linear-type functions for deriving uncertainty estimates based on indicator scores, the exponential-type approach has been mostly used in the past because of the logic that uncertainty is not increasing linearly with poorer data quality, but rather exponentially. However, no matter which kind of relationship between data quality and quantitative uncertainty estimate is used, its potentially high effect on the resulting uncertainty ranges of material flows and the inherent subjectivity owing to the lack of empirical data make it a critical aspect of the overall uncertainty analysis. Therefore, the focus needs to be rather on consistent uncertainty estimates within one MFA study than on the comparability of uncertainty estimates between different studies.

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About the Authors

David Laner is a senior researcher at the Institute for Water Quality, Resource and Waste Management (IWRWM) at the Vienna University of Technology, Vienna, Austria. **Julia Feketitsch** is a former research associate and **Helmut Rechberger** is a professor for resource management at the IWRWM. **Johann Fellner** is an assistant professor at the IWRWM and head of the Christian Doppler Laboratory for Anthropogenic Resources hosted by the IWRWM.

Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's web site:

Supporting Information S1: This supporting information contains detailed information about the characterization functions used to translate evaluation scores into uncertainty estimates, the data used to estimate plastics flows, their evaluation and uncertainty characterization, and some additional results not presented in the main article.